Climate-Aware Malaria Prediction & Explainable Healthcare AI System.

DIGITAL HEALTH AFRICA

DIGITAL HEALTH AFRICA CAPSTONE PROJECT

About Me

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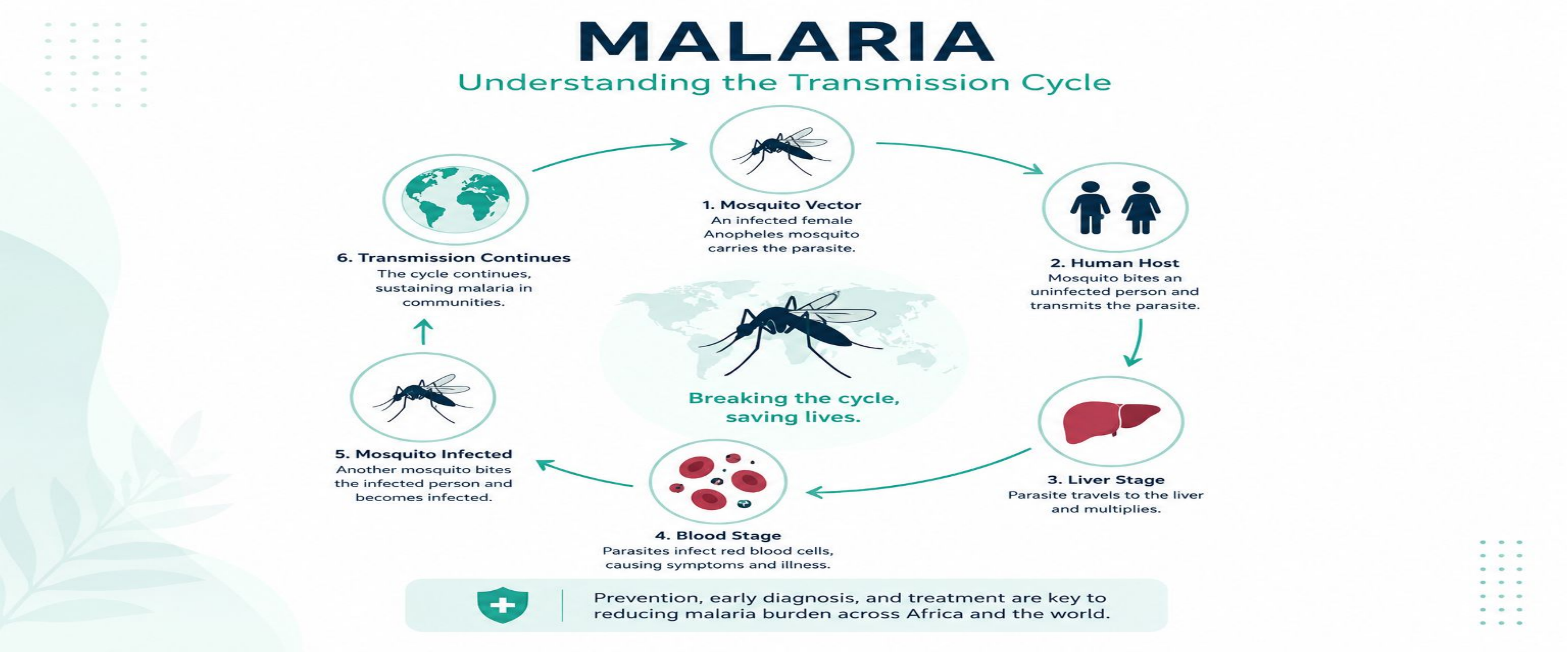
Introduction

- Malaria remains a major public health challenge in Africa, accounting for approximately 95% of global malaria cases and deaths.
- Over 249 million malaria cases and 600,000 malaria-related deaths occur globally each year, with the majority concentrated in Sub-Saharan Africa.
- Climate variability, poor sanitation, and limited healthcare access continue to influence malaria transmission across vulnerable communities.
- Advances in Artificial Intelligence (AI) and Machine Learning (ML) provide opportunities to improve malaria surveillance, forecasting, and intervention planning
- This project developed a climate-aware and spatially informed machine learning system for predicting malaria incidence across African countries using healthcare, demographic, environmental, and temporal indicators.
- The system integrates SHAP explainability and a deployed Streamlit dashboard to support malaria risk interpretation and public health decision-making.

Transmission Cycle



Problem Statement

- Malaria remains one of Africa's most significant public health challenges.
- Africa accounts for approximately 95% of global malaria cases and deaths.
- Climate variability, poor sanitation, and healthcare access influence malaria transmission.
- Traditional surveillance methods often rely on retrospective analysis.
- *Therefore, there is a need for a climate-aware and explainable machine learning system capable of integrating healthcare, demographic, environmental, temporal, and spatial indicators to improve malaria prediction and support public health decision-making across African countries.*

Project Overview

- ❑ To develop a climate-aware and explainable machine learning system for predicting malaria incidence across African countries.
- ❑ Core Components
- ❖ Climate-aware malaria forecasting
- ❖ Spatial and temporal analysis
- ❖ Explainable AI using SHAP
- ❖ Interactive healthcare dashboard
- ❖ Public health policy simulation

- ❑ Technologies Used
- ❖ Python
- ❖ Scikit-learn
- ❖ XGBoost
- ❖ SHAP
- ❖ Streamlit
- ❖ Plotly

Project Objective

Develop a climate-aware and spatially informed Machine Learning system for predicting malaria incidence across African countries using demographic, healthcare, environmental, and temporal indicators.

Dataset Summary

- The integrated climate-aware malaria dataset consisted of:
  - malaria, rainfall, and temperature datasets
  - 594 records
  - 54 African countries
  - Over 30 healthcare, demographic, climate, spatial, and engineered temporal features
- Malaria Dataset consists of healthcare indicators, demographic indicators, sanitation variables, and socio-economic indicators
- The rainfall dataset captured annual precipitation patterns across African countries,
- Rainfall range: <100 mm to >3,000 mm
- The temperature dataset contained annual mean temperatures.
- Temperature range: ~15°C to >30°C across different African climatic zones.

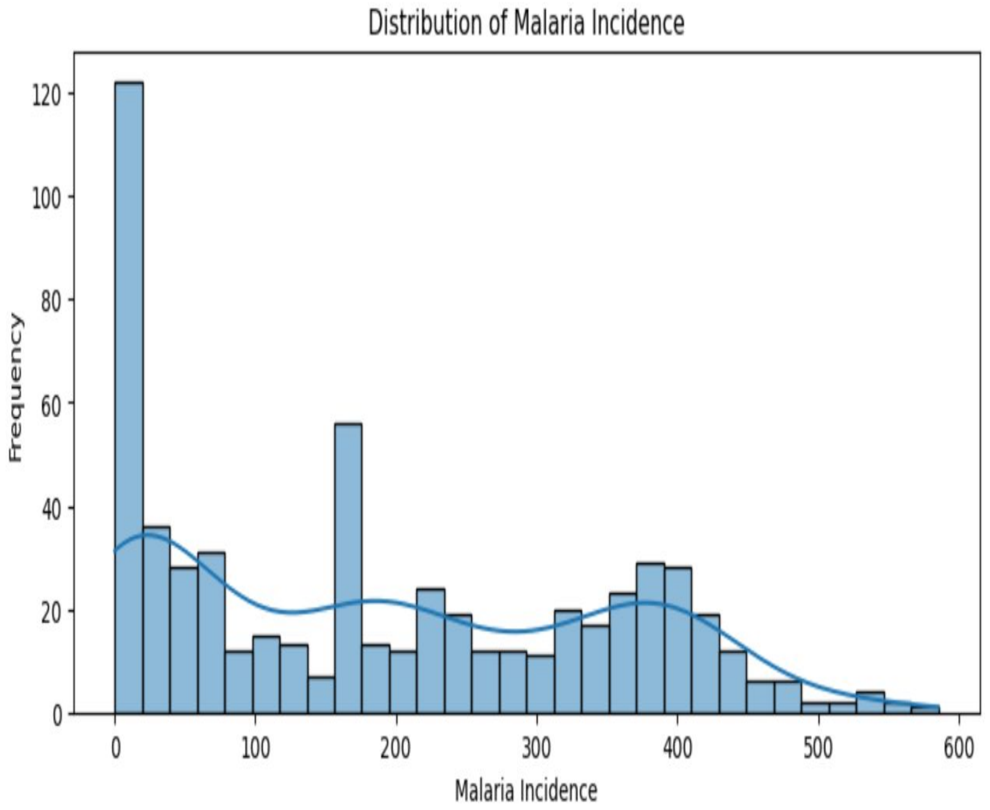
Data Preprocessing

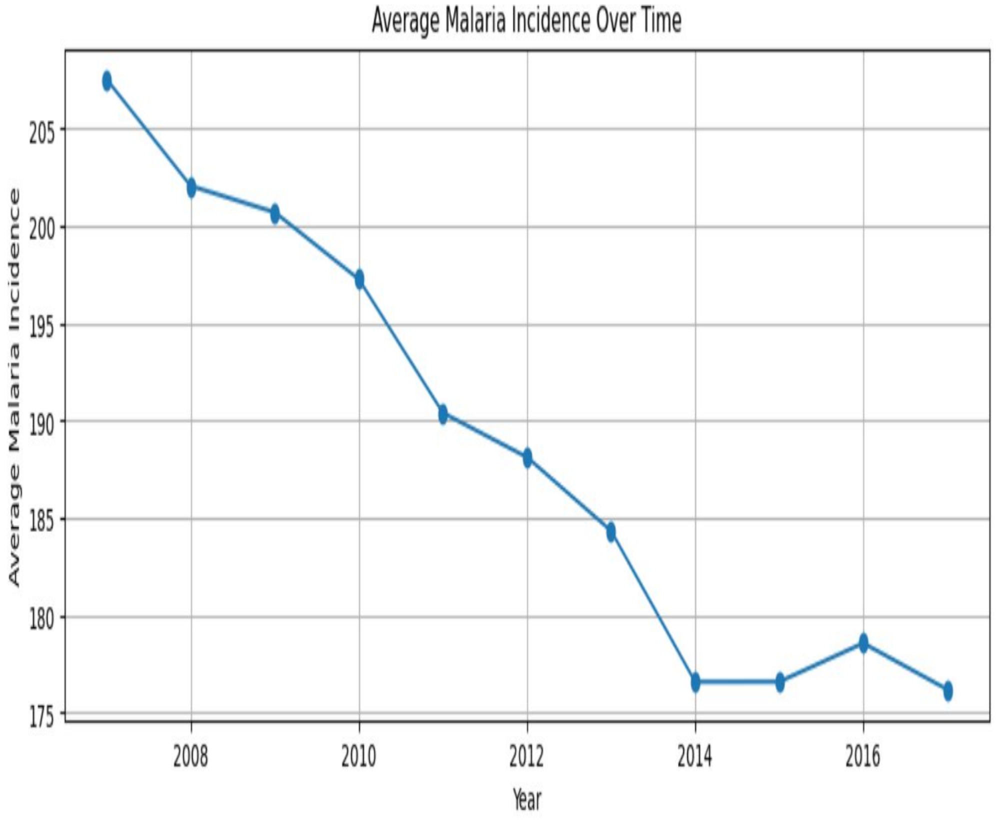
- Data preprocessing was performed to prepare the integrated malaria and climate datasets for machine learning analysis.
- Missing values were handled using appropriate imputation techniques to improve dataset quality and consistency.
- Duplicate records were identified and removed to prevent redundancy during analysis and modeling.
- Column names were standardized and data types were corrected to ensure consistent feature formatting

Data Preprocessing

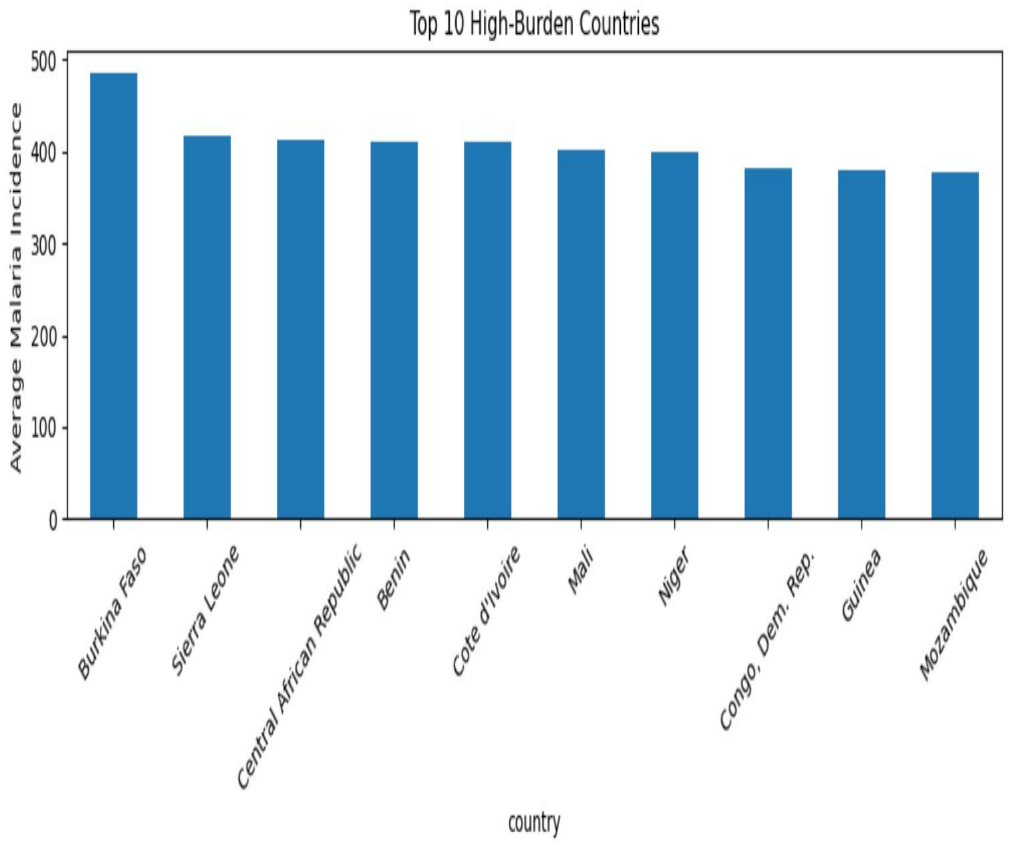
- Relevant African country records were filtered and retained for climate-aware malaria analysis.
- Rainfall and temperature datasets were transformed and merged with the malaria dataset using country and year variables.
- Feature scaling and encoding techniques were applied where necessary to prepare the data for machine learning modeling.
- The preprocessing stage produced a clean and integrated climate-aware malaria dataset suitable for exploratory analysis, feature engineering, and predictive modeling.

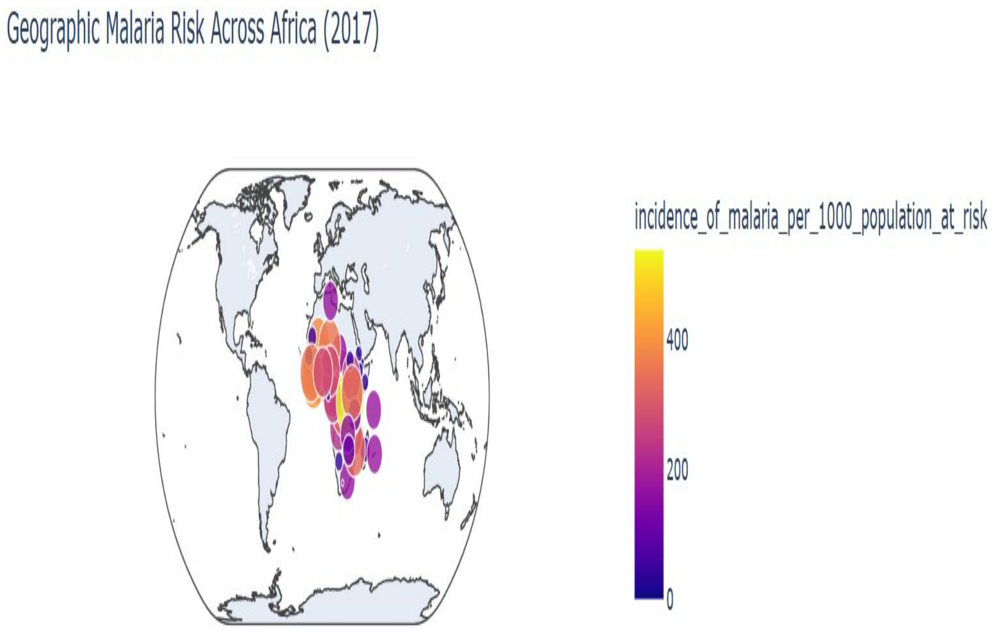
Exploratory Data Analysis (EDA)





Exploratory Data Analysis (EDA)





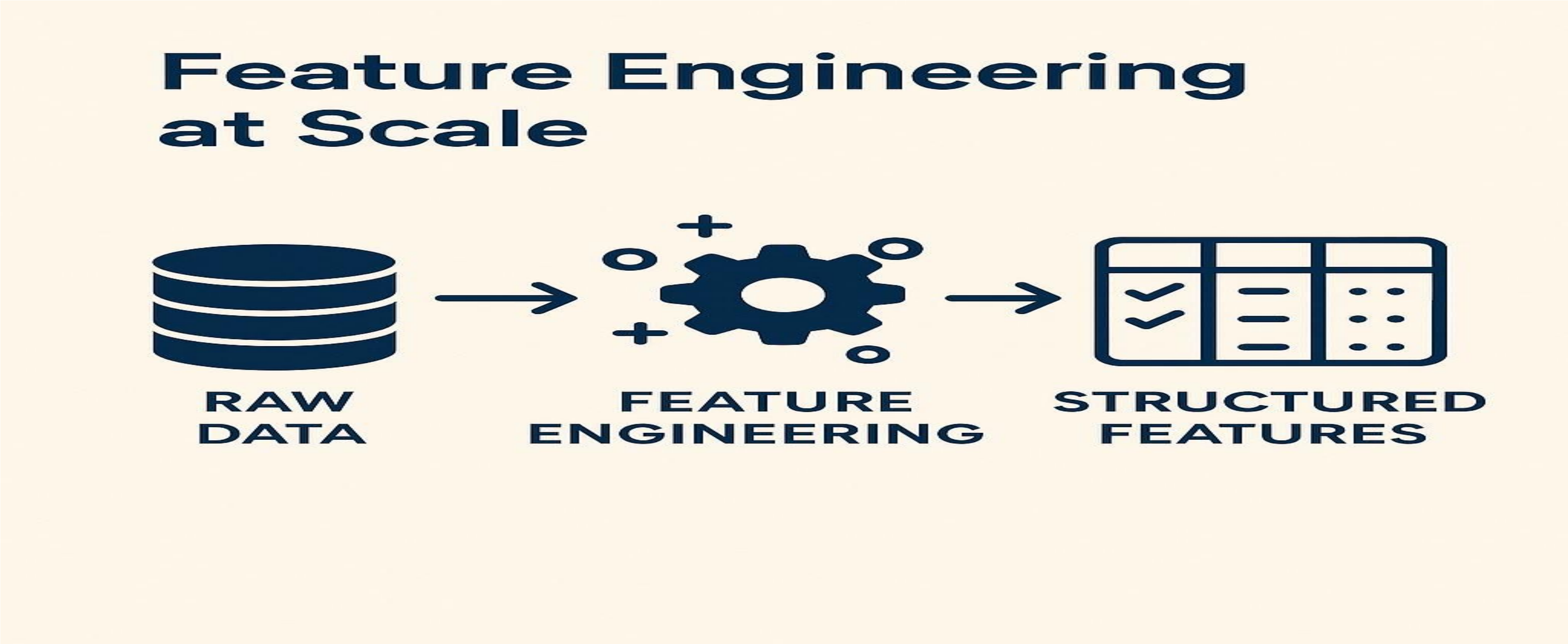
Exploratory Data Analysis (EDA)

- Exploratory Data Analysis (EDA) was conducted to examine malaria incidence patterns, climate relationships, spatial variability, and temporal trends across African countries.
- Visualization techniques including histograms, line charts, scatter plots, boxplots, correlation heatmaps, geographic risk maps, and spatial cluster visualizations were used to explore feature distributions and relationships within the integrated climate-aware malaria dataset.
- Malaria incidence varied significantly across countries, ranging from near-zero values in low-burden regions to over 580 cases per 1,000 population at risk in high-burden regions.
- Several countries consistently exhibited malaria incidence levels above 400 cases per 1,000 population at risk, highlighting substantial regional disease disparities.

Exploratory Data Analysis (EDA)

- Temporal trend analysis revealed a gradual decline in average malaria incidence from approximately 207 cases per 1,000 population at risk in 2007 to approximately 176 cases per 1,000 population at risk by 2017, representing an overall decline of nearly 15%.
- Scatter plot analysis demonstrated meaningful relationships between malaria incidence and climate-related variables such as rainfall, temperature, and climate risk index.
- Correlation heatmap analysis identified strong relationships between temporal lag features, climate variables, and malaria incidence.
- Geographic risk visualization and spatial clustering analysis revealed substantial regional malaria heterogeneity across African countries.

Feature Engineering



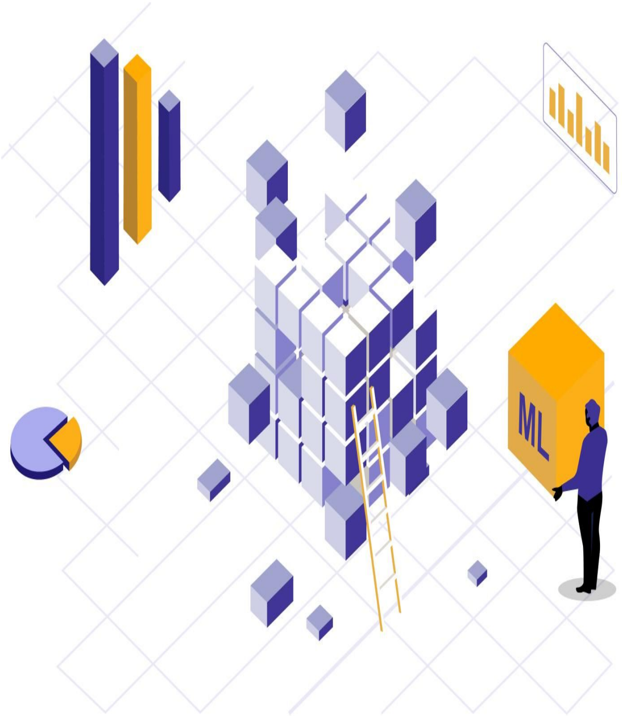
Feature Engineering

- Feature engineering was performed to improve model performance and better capture the temporal, climatic, spatial, and demographic factors influencing malaria transmission across African countries.
- One-year and two-year lag variables were created for malaria incidence, rainfall, and temperature to model delayed environmental and disease effects over time.
- Lag features were generated using only historical observations within each country grouping to reduce the risk of temporal leakage.
- Three-year rolling averages were applied to rainfall and temperature variables to capture medium-term climate exposure and accumulated environmental effects.

Feature Engineering

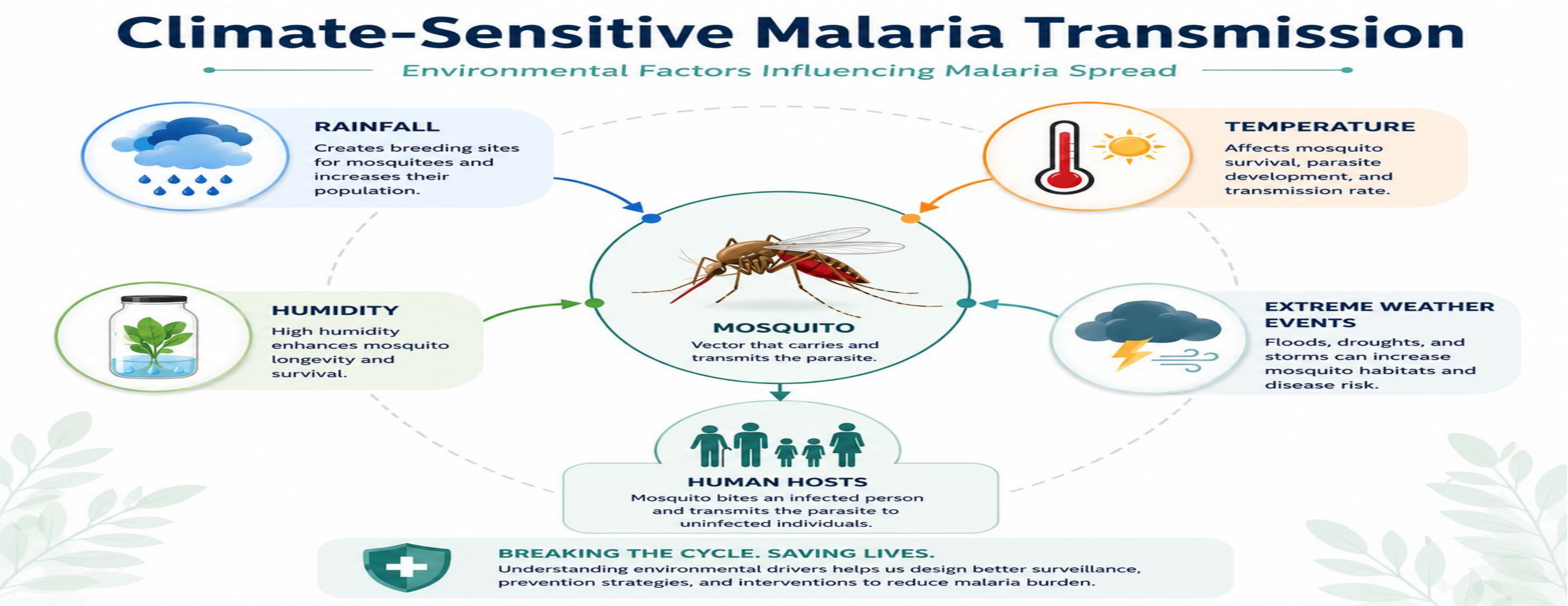
- Climate interaction variables such as climate risk index, rainfall-rural interaction, latitude-temperature interaction, and longitude-rainfall interaction were engineered to capture combined environmental and population-related influences on malaria transmission.
- Spatial feature engineering was performed using latitude and longitude variables alongside K-Means clustering to represent regional malaria heterogeneity across African countries.
- Country information was encoded into numerical form to represent country-level differences such as healthcare systems, ecological conditions, and intervention histories.
- Numerical variables including rainfall, temperature, sanitation access, and population indicators were standardized within the machine learning pipeline to ensure comparable feature scales.

ML Feature Engineering



- Overall, the engineered features improved the model's ability to capture:
  - Temporal persistence,
  - Climate-sensitive transmission dynamics, and
  - Geographic malaria variability across Africa.

Climate & Malaria Relationship



Climate Integration in This Project

- Rainfall and temperature datasets were integrated with malaria data.
- Climate-aware features such as climate risk index and rainfall interactions were engineered.
- Climate-related variables contributed significantly to malaria incidence prediction, highlighting the importance of climate intelligence in:
  - Malaria surveillance
  - Outbreak preparedness,
  - Public health intervention planning.

Machine Learning Modeling

- A production-ready machine learning pipeline was developed using Scikit-learn preprocessing and pipeline techniques to automate data transformation, feature scaling, encoding, and model training.
- TimeSeriesSplit cross-validation was used to preserve the temporal structure of the dataset and reduce data leakage during model evaluation.
- Multiple regression models were trained and compared for malaria incidence prediction, including Linear Regression,
- The models were assessed using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R^2 Score metrics.

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Model Evaluation

- Model performance was evaluated using TimeSeriesSplit cross-validation and a final holdout test dataset to preserve temporal structure, reduce data leakage, and improve model robustness.
- Ensemble-based models outperformed Linear Regression, indicating the presence of complex nonlinear relationships within the malaria prediction task.
- Among the evaluated models, the Random Forest Regressor achieved the best overall performance, with an MAE of 30.45, RMSE of 48.16, and an R^2 Score of 0.87 on the holdout test dataset.
- The relatively small dataset size of 594 records across 54 countries may limit model generalizability and increase susceptibility to overfitting, particularly for ensemble-based machine learning models.

Model Performance

The following machine learning models were evaluated:

Model	Average R^2 Score
Linear Regression	-9.90
Random Forest Regressor	0.54
Gradient Boosting Regressor	0.47
XGBoost Regressor	0.41

Final Model Performance

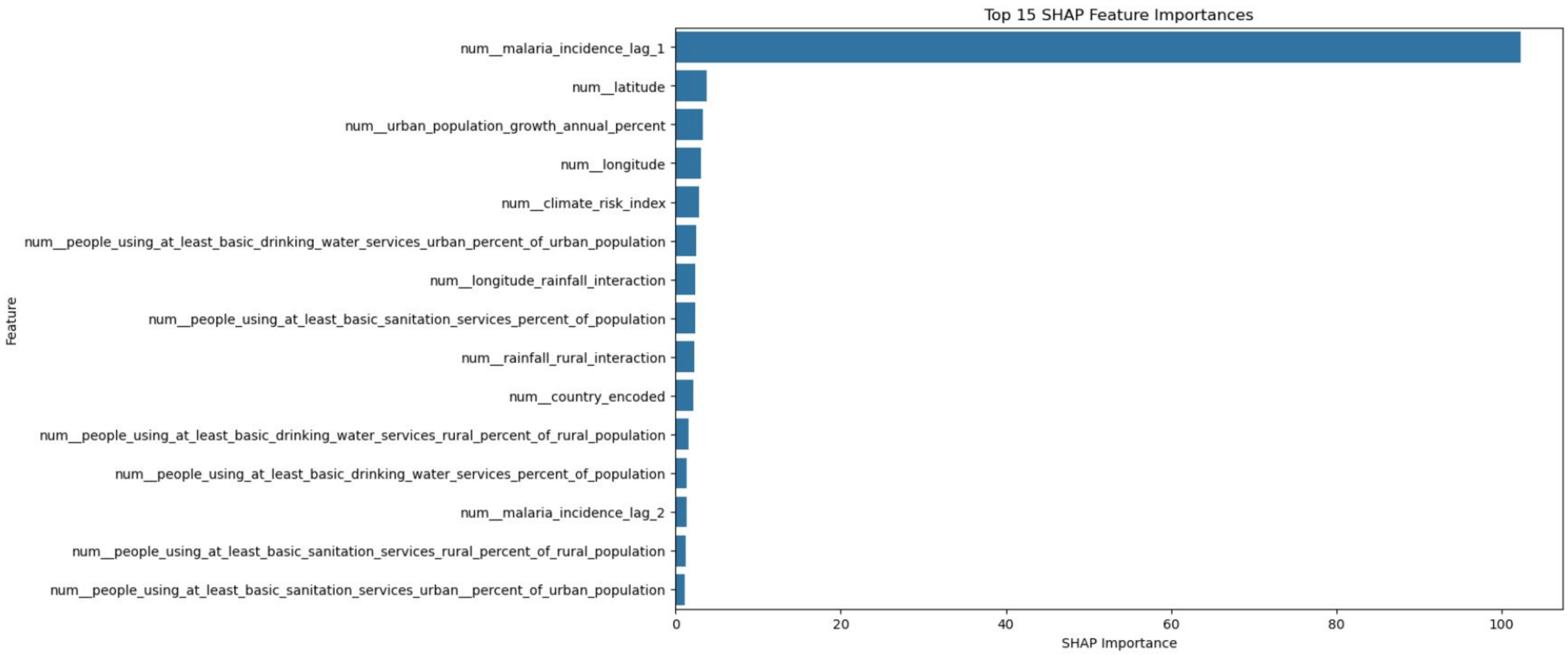
- Final holdout test evaluation demonstrated superior performance for the untuned Random Forest model:
- The untuned Random Forest model was selected as the final production-ready malaria prediction model

Model	MAE	RMSE	R^2 Score
Untuned Random Forest	30.45	48.16	0.87
Tuned Random Forest	57.36	70.54	0.73

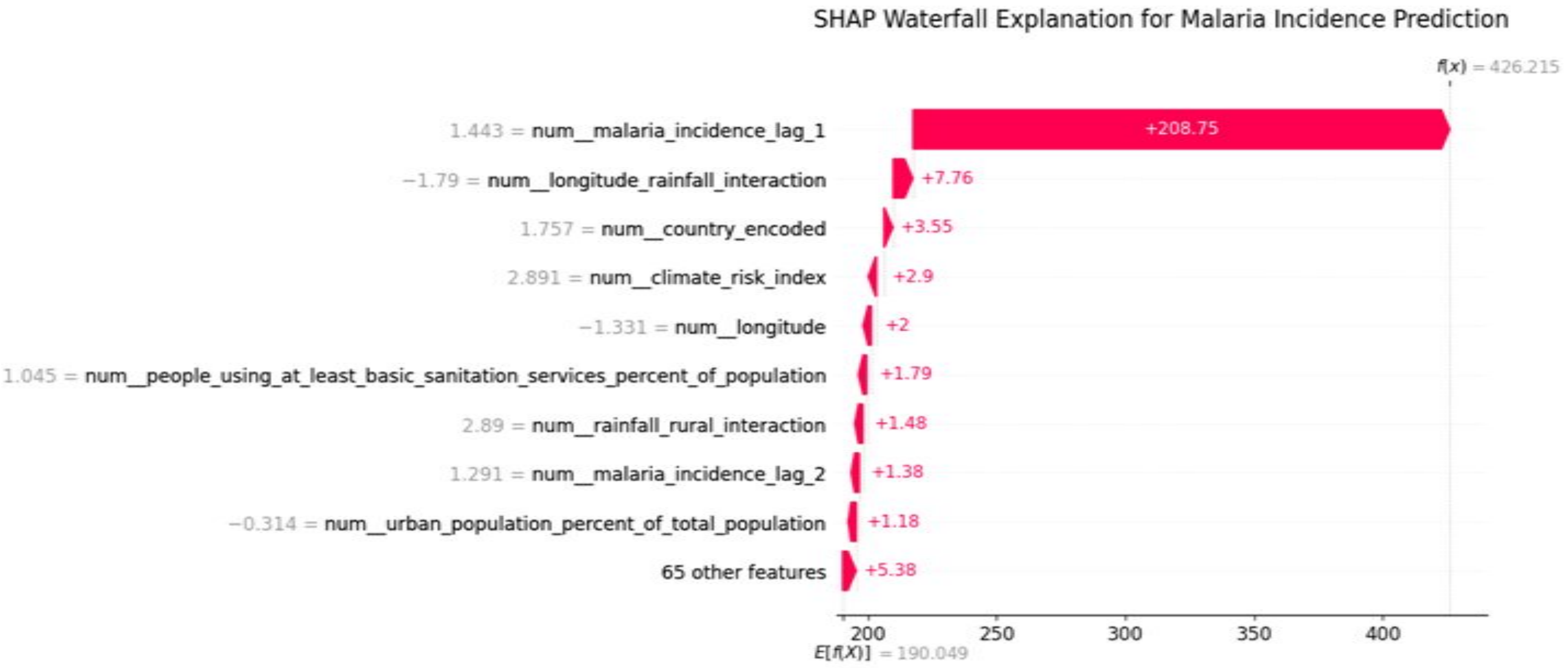
SHAP Explainability

- SHAP (SHapley Additive exPlanations) improves model transparency and identify the strongest drivers of malaria prediction.
- Previous malaria incidence was the strongest predictor of future malaria burden.
- The model demonstrated strong temporal persistence in malaria transmission dynamics.
- Climate-related variables significantly influenced prediction outcomes, including : climate risk index, rainfall interactions, temperature lag features.
- Sanitation and water access contribute to malaria vulnerability.
- Geographic and spatial heterogeneity strongly influence transmission patterns.

SHAP Explainability



Healthcare AI Interpretation



Explainable Healthcare AI.

- Predicted malaria incidence : **426.22** cases per 1,000 population at risk.
- Baseline average prediction:**190.05** cases per 1,000 population at risk
- Strongest Predictor: Previous Malaria Incidence (malaria_incidence_lag_1)Contribution:**+208.75** SHAP units
- Demonstrated strong temporal persistence in malaria transmission.

- Additional Contributing Factors:
- Longitude-rainfall interaction**: +7.76**
- Country encoding**: +3.55**
- Climate risk index**: +2.90**
- Longitude**: +2.00**
- Sanitation access**: +1.79**
- Rainfall-rural interaction**: +1.48**

Key Impact

- ***SHAP analysis improved model interpretability and strengthened the system's applicability for :***
  - ***Healthcare decision support,***
  - ***Malaria surveillance,***
  - ***Intervention planning, &***
  - ***Public health policy development.***

What-If Policy Simulation

- ❑ Evaluate the potential impact of improving sanitation access on malaria incidence predictions.
- ❑ Simulation Approach: Increased access to basic sanitation services for selected country-year observations.
- ❑ Generated updated malaria incidence predictions using the trained Random Forest model.
- ❑ Example Scenario: Country: Mozambique -- Year2008
- ❑ Baseline Prediction: 399.79 cases per 1,000 population at risk.
- ❑ Intervention Scenario: 20% increase in sanitation access.
- ❑ Predicted Incidence After Intervention: 399.19 cases per 1,000 population at risk.
- ❑ Estimated Reduction: 0.60 cases per 1,000 population at risk. Similar intervention scenarios produced reductions ranging from:0.60–0.71 cases per 1,000 population at risk.

Streamlit Dashboard Deployment

- A production-ready Streamlit dashboard was successfully deployed publicly to support:
  - Climate-aware malaria prediction
  - Interactive healthcare analytics
  - Real-time decision support
- The deployed dashboard integrates: Machine Learning, Climate Intelligence, Explainable AI, Spatial Analytics, & Public Health Decision Support
- The system demonstrates how AI-driven healthcare analytics can support malaria surveillance, intervention planning, climate-sensitive outbreak monitoring, evidence-based public health policy.

Policy Recommendations

- Strengthen malaria surveillance in historically high-burden regions to improve early detection and outbreak preparedness.
- Integrate rainfall, temperature, and climate-risk monitoring into malaria early-warning systems.
- Improve sanitation and drinking water access in vulnerable communities to reduce environmental and infrastructure-related malaria risk.
- Prioritize rural and climate-sensitive regions for targeted intervention programs and healthcare resource allocation.
- Utilize geographic risk mapping and predictive analytics to support data-driven public health decision-making.
- Apply explainable AI techniques to improve transparency and interpretation of healthcare prediction systems for policymakers and healthcare stakeholders.

Ethical Considerations

- Model outputs are predictive, not causal and Predictions should not replace healthcare professionals or epidemiological expertise.
- Variations in surveillance quality, testing availability, and healthcare infrastructure may introduce bias into predictions and underreporting may affect malaria burden estimation in some regions.
- Explainable AISHAP explainability was incorporated to improve transparency and support responsible healthcare interpretation.
- The Streamlit dashboard was designed as a decision-support and exploratory analytics tool, not an automated intervention system.
- Responsible deployment of healthcare AI requires ongoing validation, monitoring, contextual interpretation, and healthcare stakeholder involvement.

Limitations

- Some African regions had incomplete climate and healthcare records.
- Country-level aggregation may not fully capture local transmission dynamics.
- The model relies on reported malaria incidence, which may underestimate actual disease burden in areas with limited testing infrastructure.
- Additional environmental variables such as humidity and vegetation indices were not included due to data availability constraints.
- Streamlit deployment environment differences introduced minor reproducibility considerations during online deployment.
- ***Despite these limitations, the model demonstrated strong predictive capability and meaningful public health interpretability.***

Potential Future Work

- Several opportunities exist for extending and improving the current climate-aware malaria prediction system in future studies.
- Explore advanced deep learning and spatiotemporal forecasting approaches such as LSTM networks, Temporal Fusion Transformers, Graph Neural Networks (GNNs), and hybrid climate-health forecasting models.
- Extend the dashboard with geospatial drill-down capabilities, regional forecasting modules, and automated outbreak-alert systems for healthcare stakeholders and national malaria control programs.
- Integrate Large Language Models (LLMs) and conversational AI interfaces to support interactive public health analytics and healthcare stakeholder interpretation.
- Explore mobile and low-resource deployment strategies to improve accessibility of climate-aware healthcare AI systems within underserved regions across Africa.

Conclusion

- A climate-aware and spatially informed machine learning system was successfully developed for malaria incidence prediction across African countries.
- Random Forest achieved the best predictive performance with an R^2 Score of 0.87.SHAP explainability improved model transparency and interpretation.
- The deployed Streamlit dashboard demonstrated real-world healthcare AI deployment capability.
- The project demonstrates the potential of explainable healthcare AI systems for climate-sensitive malaria surveillance, intervention planning, and public health decision-making across Africa.

Live Dashboard

Public App Link:
https://malaria-app-app-hp7r78ub8agnjku4n9stpf.streamlit.app/

GitHub Repository:
https://github.com/drsam-israel/malaria-streamlit-app

Thank You

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